

Continuous functions – 8

Sets of discontinuity, oscillation and the Baire category theorem

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Theoretical minimum.

Theorem 1 (structure of the discontinuity set). For any $f: \mathbb{R} \rightarrow \mathbb{R}$ the set $\{x : \text{osc } f(x) \geq \varepsilon\}$ is closed; hence the set of points of discontinuity is an F_σ set, and the set of points of continuity is a G_δ set.

Theorem 2 (Baire). A complete metric space is not a countable union of nowhere dense sets. Consequently $\mathbb{Q}$ is not a G_δ subset of $\mathbb{R}$, so no function is continuous exactly at the rational points, whereas Thomae's function is continuous exactly at the irrational ones.

Theorem 3 (Lebesgue's criterion). A bounded function on $[a, b]$ is Riemann integrable if and only if its set of points of discontinuity has measure zero.

1. Prove Theorem 1, and deduce that the set of continuity points of any function is a G_δ set. Then prove that every F_σ set is the discontinuity set of some function.
2. (S.T. Yau CSMC 2014, team) Construct an increasing function on $\mathbb{R}$ whose set of points of discontinuity is exactly $\mathbb{Q}$.
3. Prove Baire's theorem and deduce that no function on $\mathbb{R}$ is continuous exactly at the rational points; deduce also that a pointwise limit of continuous functions has a dense set of continuity points.
4. (S.T. Yau CSMC 2016) Prove Lebesgue's criterion: a bounded function on $[a, b]$ is Riemann integrable if and only if its set of discontinuities has measure zero.

5. (RMO) Let $f, g, h: \mathbb{R} \rightarrow \mathbb{R}$ be such that f is differentiable, g and h are monotone and $f' = f + g + h$. Prove that the sets of discontinuity points of g and of h coincide.